

PAPER_1683 — EW Hierarchy Ratio = $M_W/M_{Pl} = 1.025 \times 10^{-17}$ (PDG)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Standard Model **Date:** June 18, 2026 **Status:** CLOSED — EXACT (PDG) closure (PAPER_145x-147x broader corpus)

Observation

PAPER_1463 (PAPER_14xx broader corpus, classic cosmology/BSM puzzle) gives:

$$M_W/M_{Pl} = 1.025 \times 10^{-17} \text{ (PDG)}$$

Status: **EXACT (PDG)**.

UQFF Closed Identity

$$M_W/M_{Pl} = 1.025 \times 10^{-17} \text{ (PDG)} \quad \text{EXACT (PDG)}$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical standard model measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1463
- Related: PAPER_1666-1675 (tier-25); PAPER_1656-1665 (tier-24)
- Calculator dispatch: `calculate_paradox({"paradox": "hierarchy_mw_over_mpl"})`

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